

Heat-content maximization on compact metric graphs at every time

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September 16, 2026

Abstract

Among finite compact connected metric graphs of prescribed total length with a nonempty Dirichlet vertex set and standard conditions elsewhere, the interval with one Dirichlet and one Neumann endpoint maximizes heat content at every positive time. Equality at a single positive time characterizes this interval, up to subdivision. This answers the all-times question of Bifulco and Täufer. We prove the stronger statement that heat evolution on any such graph is dominated in cumulative decreasing rearrangement by mixed-boundary interval evolution of the rearranged initial datum. The argument combines the classical metric-graph level-set inequality with the zero-order elliptic concentration comparison and implicit-time iteration of parabolic symmetrization. The rigidity proof uses the half-time energy identity and strict edgewise concavity of the heat evolution of the constant function. No arithmetic condition on the edge lengths is imposed. We also prove that, among graphs of fixed length carrying a finite nonnegative killing measure of fixed positive mass, the interval with all killing at one endpoint maximizes heat content at every time. This comparison allows singular measures and arbitrary nonnegative initial data, and equality at a single positive time characterizes the interval and its endpoint measure. For the same general killing measures we prove sharp Kohler-Jobin inequalities at fixed length and total killing, and at fixed torsion and total killing. A ground-state distribution estimate controls the potential terms and compares the resulting scalar differential inequality with the cosine profile; both equality cases characterize the endpoint-killed interval. The stationary joint comparisons also extend to the nonlinear p -Laplacian for every $1 < p < \infty$, with explicit finite-strength constants and the same rigidity.

Keywords: metric graph; heat content; rearrangement; concentration comparison; Faber-Krahn inequality; rigidity.

Mathematics Subject Classification (2020): 35K05, 34B45, 49Q10.

1. Introduction and main results

Let Γ be a finite compact connected metric graph, with positive edge lengths and total length $L > 0$. Let D be a nonempty subset of its vertices. We assume that $\Gamma \setminus D$ is connected. The Laplacian has Dirichlet conditions at D and standard continuity and Kirchhoff conditions at every other vertex. A standard condition at a vertex of degree one is a Neumann condition. Loops and multiple edges are permitted; degrees and Kirchhoff sums count edge incidences, so a loop contributes twice.

Write A_Γ for the nonnegative realization of $-d^2/dx^2$ and $S_\Gamma(t) = e^{-tA_\Gamma}$. Its heat content is

$$Q_\Gamma(t) = \int_\Gamma S_\Gamma(t) 1 \, dx. \tag{1}$$

We compare Γ to $I_L = (0, L)$, with Neumann condition at 0 and Dirichlet condition at L . Denote the corresponding operator, semigroup and heat content by A_I , $S_I(t)$ and $Q_I(t)$.

Theorem 1.1 (heat-content maximization and rigidity). For every graph described above and every $t > 0$,

$$Q_\Gamma(t) \leq Q_I(t). \quad (2)$$

Equality for one positive time holds if and only if Γ is a path of length L , D consists of one endpoint, and all other vertices have standard conditions. Equivalently, after suppressing vertices of degree two, the graph is the Dirichlet-Neumann interval. In particular, for every other admissible graph the inequality is strict at every positive time.

For a nonnegative measurable function f on a space of length L , let f^* be its decreasing rearrangement on $(0, L)$, and set

$$\mathcal{C}_f(s) = \int_0^s f^*(r) dr, \quad 0 \leq s \leq L. \quad (3)$$

We write $f \preceq g$ if $\mathcal{C}_f(s) \leq \mathcal{C}_g(s)$ for all s . Equality of total integrals is not part of this definition. Thus the order is weak submajorization, also called concentration order in the parabolic comparison literature.

Theorem 1.2 (concentration comparison). Let $0 \leq f \in L^2(\Gamma)$ and let $0 \leq g \in L^2(0, L)$ be nonincreasing. If

$$\mathcal{C}_f(s) \leq \int_0^s g(r) dr \quad \text{for every } s \in [0, L], \quad (4)$$

then, for every $t > 0$,

$$\mathcal{C}_{S_\Gamma(t)f}(s) \leq \int_0^s S_I(t)g(r) dr \quad \text{for every } s \in [0, L]. \quad (5)$$

The interval function on the right is nonnegative and nonincreasing. In particular, one may take $g = f^*$.

Taking $f = g = 1$ and $s = L$ proves the inequality in Theorem 1.1. Its one-time equality assertion needs an additional argument and is proved in Section 5.

1.1. Relation to earlier work

Bifulco and Täufer [BT, Theorem 2.6] prove the interval comparison for sufficiently large times, and for sufficiently small times when the edge lengths are rationally dependent. Their final Remark 2.9 leaves the comparison at every positive time open. Theorem 1.1 resolves that question without a restriction on edge lengths or on the time. The graph, vertex conditions and total-length normalization here agree with their Definitions 2.1 and 2.2. Their comparison interval is written with its Dirichlet endpoint at 0; reflection gives our convention.

Bifulco and Mugnolo [BM] derive a heat-content formula, short-time asymptotics and surgery principles for compact quantum graphs. Their Section 6 includes comparisons under imposing Dirichlet conditions, taking subgraphs and attaching pendant graphs, as well as heat-content-preserving constructions. These are a different set of comparisons from (5), which rearranges arbitrary initial data and compares graphs of the same total length. We use the ordinary Kirchhoff heat semigroup studied in that work.

The geometric rearrangement input is classical. In the proof of [F, Lemma 3], Friedlander uses the number of points in a regular level set to compare the graph Dirichlet energy to the energy of a monotone rearrangement on an interval. That proof also explains how equality in level multiplicity excludes branching. We give the relevant calculation because the zero-order term of the resolvent equation must be retained.

The passage from elliptic concentration comparison to parabolic comparison by implicit-time discretization is likewise classical. Vázquez [V, Sections 5 and 7] develops precisely this strategy for Euclidean equations; see in particular the zero-order comparison and the parabolic iteration in the author's available text. We adapt that mechanism to the mixed-boundary graph problem. Neither the level-set Cauchy-Schwarz inequality nor implicit Euler iteration is asserted to be new. The contribution is the all-times graph comparison (5), its answer to [BT, Remark 2.9], and the one-time rigidity theorem. The elementary consequences in Section 6 are included as applications of the same result. Section 7 extends the concentration theorem to nonnegative killing measures of prescribed total mass, using a Robin interval comparator and a boundary inequality for cumulative resolvents. It includes delta-vertex conditions, nonnegative potentials and finite singular measures.

2. Graph forms and rearrangement preliminaries

The measure on Γ is the sum of Lebesgue measures on the edges; vertices have measure zero. The closed quadratic form of $\mathbf{a}_\Gamma$ is

$$\mathbf{a}_\Gamma[u] = \sum_e \int_0^{\ell_e} |u'_e(x)|^2 dx, \quad \mathcal{D}(\mathbf{a}_\Gamma) = \{u \in H^1(\Gamma) : u|_D = 0\}. \quad (6)$$

Here $H^1(\Gamma)$ consists of edgewise H^1 functions continuous at the vertices. The operator domain consists of edgewise H^2 functions in the form domain satisfying the Kirchhoff sum at each vertex outside D . The same construction on I_L , with form domain $\{v \in H^1(0, L) : v(L) = 0\}$, gives A_I .

Both operators are nonnegative and self-adjoint, with compact resolvent. They are strictly positive: if $p \in D$, integration along a simple path from p to any point gives

$$|u(x)|^2 \leq L \mathbf{a}_\Gamma[u], \quad \|u\|_2^2 \leq L^2 \mathbf{a}_\Gamma[u]. \quad (7)$$

The truncations $u \mapsto u_+$ and $u \mapsto \min\{1, \max\{u, 0\}\}$ preserve the form domain and do not increase its energy. The resolvents and semigroups are therefore positivity preserving and sub-Markovian. In particular,

$$0 \leq S_\Gamma(t)1 \leq 1, \quad 0 \leq S_I(t)1 \leq 1. \quad (8)$$

We recall two elementary rearrangement facts. For nonnegative $f \in L^1$ on a nonatomic space of length L ,

$$\mathcal{C}_f(s) = \sup_{|E|=s} \int_E f, \quad \sup_{0 \leq s \leq L} |\mathcal{C}_f(s) - \mathcal{C}_h(s)| \leq \|f - h\|_1. \quad (9)$$

The supremum formula follows by filling a superlevel set of f and, if necessary, part of one level set. The second assertion follows by comparing the integrals on each candidate set. Also,

$$\|f^* - h^*\|_{L^2(0, L)} \leq \|f - h\|_{L^2(\Gamma)} \quad (10)$$

for nonnegative f, h . Indeed, Hardy-Littlewood gives $\int_{\Gamma} fh \leq \int_0^L f^* h^*$, while rearrangement preserves the two squared norms. Expanding the squared distances proves (10).

Lemma 2.1 (a Lipschitz rearrangement). Suppose $u \geq 0$ is continuous on Γ , vanishes at D , and is Lipschitz on every edge, with maximum edge Lipschitz constant K . Then u^* has a Lipschitz representative on $[0, L]$ with Lipschitz constant at most K , and $u^*(L) = 0$.

Proof. The case $K = 0$ is immediate. Choose a maximum point and a simple path from it to D . For $0 \leq c_1 < c_2 \leq \max u$, this path contains a subarc along which u passes from c_2 to c_1 while remaining between those values. The length of that subarc is at least $(c_2 - c_1)/K$. The path is simple, so its length is counted without multiplicity in the graph measure. Consequently

$$|\{c_1 < u < c_2\}| \geq \frac{c_2 - c_1}{K}. \quad (11)$$

This inverse-distribution estimate gives the Lipschitz assertion; jumps of the distribution due to plateaus correspond to constant portions of the rearrangement and cause no difficulty. Finally, continuity at a Dirichlet vertex shows that the essential infimum is zero. Hence the representative has $u^*(L) = 0$. $\square$

3. Resolvent concentration comparison

For $h > 0$, put

$$R_h = (1 + hA_{\Gamma})^{-1}, \quad B_h = (1 + hA_I)^{-1}. \quad (12)$$

Lemma 3.1 (the interval resolvent). If $g \geq 0$ is nonincreasing and belongs to $L^2(0, L)$, then $v = B_h g$ is nonnegative and nonincreasing. With $V(s) = \int_0^s v$ and $G(s) = \int_0^s g$,

$$-hV'' + V = G, \quad V(0) = 0, \quad V'(L) = 0. \quad (13)$$

If g_1, g_2 are two such functions and $\int_0^s g_1 \leq \int_0^s g_2$ for all s , then $\int_0^s B_h g_1 \leq \int_0^s B_h g_2$ for all s .

Proof. First let g be smooth on $[0, L]$. The equation is $-hv'' + v = g$, with $v'(0) = 0$ and $v(L) = 0$. Positivity of the resolvent gives $v \geq 0$, hence $v'(L) \leq 0$. Its derivative $q = v'$ satisfies

$$-hq'' + q = g' \leq 0, \quad q(0) = 0, \quad q(L) \leq 0. \quad (14)$$

Testing against q_+ gives $q \leq 0$. Every nonnegative nonincreasing L^2 function can be approximated in L^2 by smooth nonnegative nonincreasing functions: first truncate, then extend by constant endpoint values and mollify. Boundedness of B_h on L^2 passes the conclusion to the limit; the cone of nonincreasing functions is closed in L^2 .

Integration of the resolvent equation from 0 to s , using $v'(0) = 0$, proves (13). This remains valid for L^2 data because $v \in H^2$. For the last assertion, the difference $W = V_1 - V_2$ satisfies $-hW'' + W \leq 0$, $W(0) = 0$ and $W'(L) = 0$. Thus

$$h \int_0^L |(W_+)'|^2 + \int_0^L |W_+|^2 \leq 0, \quad (15)$$

and $W \leq 0$. $\square$

Lemma 3.2 (graph resolvent comparison). For every $0 \leq f \in L^2(\Gamma)$,

$$\mathcal{C}_{R_h f}(s) \leq \int_0^s B_h(f^*)(r) dr \quad (0 \leq s \leq L). \quad (16)$$

Consequently, if g satisfies the hypotheses and concentration bound of Theorem 1.2, then

$$\mathcal{C}_{R_h f}(s) \leq \int_0^s B_h g(r) dr. \quad (17)$$

Proof. We first suppose that on every closed edge the load f is a strictly positive nonconstant polynomial. Loads need not be continuous across vertices. The solution $u = R_h f$ satisfies

$$-hu_e'' + u_e = f_e \quad (18)$$

on each edge, so it extends analytically past both endpoints as an edgewise function. It is strictly positive outside D . For completeness, a zero interior minimum contradicts (18) and $f_e > 0$. At a non-Dirichlet vertex with value zero, all outgoing derivatives would be nonnegative, and their Kirchhoff sum would make them all zero. Equation (18) would then give a negative second derivative into each incident edge at that endpoint, contradicting nonnegativity. This proves the positivity assertion.

The function u cannot be constant on an edge, because (18) would force f to be constant there. Each edge therefore has only finitely many critical points: its derivative is analytic on a neighborhood of the closed edge and is not identically zero. Exclude their values and all vertex values. For every remaining $c \in (0, \max u)$, write

$$E_c = \{u > c\}, \quad \mu(c) = |E_c|, \quad N(c) = \#\{u = c\}, \quad a(c) = \sum_{u=c} |u'|, \quad b(c) = \sum_{u=c} \frac{1}{|u'|}. \quad (19)$$

All sums run over interior edge points. Continuity along a path from a maximum to D gives $N(c) \geq 1$. Define $U = \mathcal{C}_u$ and $F = \mathcal{C}_f$. Integration of (18) over E_c gives

$$ha(c) + U(\mu(c)) = \int_{E_c} f \leq F(\mu(c)). \quad (20)$$

To check the sign and vertex terms, the derivative pointing out of a superlevel interval is $-|u'|$ at a regular boundary point. At a vertex contained in E_c , continuity puts all incident edge germs in E_c , and their fluxes cancel by Kirchhoff. No Dirichlet vertex belongs to E_c . These observations give exactly the positive flux term on the left of (20).

The one-dimensional coarea formula and inverse differentiation give

$$-\mu'(c) = b(c), \quad U''(\mu(c)) = (u^*)'(\mu(c)) = -\frac{1}{b(c)}. \quad (21)$$

Cauchy-Schwarz, in the form already used for graph rearrangement in [F], yields

$$a(c)b(c) \geq N(c)^2 \geq 1. \quad (22)$$

Combining (20)-(22), we obtain

$$-hU'' + U \leq F. \quad (23)$$

There is no omitted singular part in (23). The function u^* is Lipschitz by Lemma 2.1, so $U \in W^{2,\infty}(0, L)$. There are no plateaus of u , and hence the finitely many exceptional levels yield only finitely

many exceptional ranks $\mu(c)$. Thus the displayed inequality holds almost everywhere and in the weak sense. Its boundary data are

$$U(0) = 0, \quad U'(L) = 0. \quad (24)$$

By Lemma 3.1, the cumulative integral V of $B_h(f^*)$ satisfies the equality in (23), with the same boundary data. The positive-part test (15) proves $U \leq V$.

To remove the polynomial assumption, approximate an arbitrary nonnegative L^2 load by strictly positive nonconstant polynomials on every closed edge. Such loads are dense: separately on each edge, approximate first by a nonnegative continuous function, then uniformly by a polynomial, adding a vanishing positive constant to ensure strict positivity. A vanishing linear perturbation makes a constant polynomial nonconstant while preserving positivity. There are only finitely many edges.

For an approximating sequence $f_n \rightarrow f$ in L^2 , resolvent boundedness and (10) give

$$R_h f_n \rightarrow R_h f, \quad B_h(f_n^*) \rightarrow B_h(f^*) \quad \text{in } L^2. \quad (25)$$

Finite total length converts these to L^1 convergences, and (9) passes the cumulative inequality to the limit. This proves (16). Finally apply Lemma 3.1 to the decreasing interval data f^* and g to obtain (17). $\square$

The comparison retains the zero-order term in (18). Dropping that term would give a Poisson comparison, which does not by itself iterate to the required heat inequality.

4. Passage to the heat semigroup

Proof of Theorem 1.2. Fix $h > 0$ and define

$$u_0 = f, \quad u_j = R_h u_{j-1}, \quad v_0 = g, \quad v_j = B_h v_{j-1}. \quad (26)$$

All these functions are nonnegative, and every v_j is nonincreasing by Lemma 3.1. The hypothesis at $j = 0$ and (17) give inductively

$$\mathcal{C}_{u_j}(s) \leq \int_0^s v_j(r) dr \quad (0 \leq s \leq L, j \geq 0). \quad (27)$$

For a fixed $t > 0$, take $h = t/n$ and $j = n$. The spectral theorem gives

$$\left(1 + \frac{t}{n} A_\Gamma\right)^{-n} f \longrightarrow S_\Gamma(t)f, \quad \left(1 + \frac{t}{n} A_I\right)^{-n} g \longrightarrow S_I(t)g \quad \text{in } L^2. \quad (28)$$

Indeed, $(1 + t\lambda/n)^{-n} \rightarrow e^{-t\lambda}$ for every $\lambda \geq 0$, and both multipliers are bounded by one. Dominated convergence in the spectral measure proves (28). Passing to the limit in (27) by (9) proves (5), uniformly in s . Closedness of the cone of nonincreasing functions gives the remaining monotonicity assertion. $\square$

This is the classical implicit-time strategy of [V], with the graph resolvent inequality supplied by Section 3. In particular, no differentiability of a time-dependent rearrangement and no assumption about regular heat levels are needed to prove the inequality.

5. Equality at a single time

The proof of rigidity uses the constant initial datum. Put

$$w(t) = S_\Gamma(t)1, \quad z(t) = S_I(t)1. \quad (29)$$

We first record a strict regularity property of this particular heat evolution.

Lemma 5.1 (strict edgewise concavity). For every $t > 0$, the function $w(t)$ is smooth on the closure of each individual edge, positive outside D , and

$$-w_e''(t, x) = A_\Gamma w(t, x) > 0 \quad (0 < x < \ell_e). \quad (30)$$

Consequently, each open edge has at most one critical point of $w(t)$ and has no constant subinterval.

Proof. Spectral smoothing gives $w(t) \in \mathcal{D}(A_\Gamma^m)$ for every integer $m \geq 1$. Repeated use of the edge equation for the operator gives edgewise H^{2m} regularity, and hence smoothness up to each endpoint. On a graph connected outside its Dirichlet set, the Kirchhoff heat semigroup is positivity improving on $\Gamma \setminus D$.

Here is a maximum-principle justification of the positivity statement used in this proof. For nonzero nonnegative initial data, positivity preservation and injectivity of $S_\Gamma(t)$ give a nonzero nonnegative profile at positive times. An interior zero at a positive time forces the solution to vanish backwards on that edge by the parabolic strong maximum principle. At a standard vertex where the nonnegative solution has value zero, all outgoing derivatives are nonnegative. Their sum is zero, so every one vanishes; the parabolic boundary point lemma excludes a positive incident edge with zero outgoing derivative. Thus positivity propagates across each standard vertex. Connectivity of $\Gamma \setminus D$ gives positivity throughout the open edges and at its standard vertices. This is the usual strict-positivity property of the connected graph heat kernel; see also the discussion in [BM, Section 1].

Sub-Markovianity and positivity give

$$w(t+h) = S_\Gamma(t)S_\Gamma(h)1 \leq S_\Gamma(t)1 = w(t) \quad (h > 0). \quad (31)$$

Differentiation at positive time in L^2 implies $A_\Gamma w(t) = -\partial_t w(t) \geq 0$. It is not the zero function: otherwise (7) would give $w(t) = 0$, which contradicts positivity. Commutation through the semigroup yields

$$A_\Gamma w(t) = S_\Gamma(t/2)A_\Gamma w(t/2). \quad (32)$$

Positivity improvement applied to the nonzero nonnegative datum on the right proves (30). The final assertion follows because w_e' is strictly decreasing. $\square$

Lemma 5.2 (equal norms under concentration). Suppose $a, b \geq 0$ belong to L^2 on spaces of the same finite measure, $a \preccurlyeq b$, and $\|a\|_2 = \|b\|_2$. Then $a^* = b^*$ almost everywhere.

Proof. For $c \geq 0$ define

$$H_a(c) = \int (a - c)_+ = \sup_{0 \leq s \leq L} \{\mathcal{C}_a(s) - cs\}. \quad (33)$$

The same formula holds for b , so $H_a \leq H_b$. Tonelli gives

$$\|a\|_2^2 = 2 \int_0^\infty H_a(c) dc, \quad \|b\|_2^2 = 2 \int_0^\infty H_b(c) dc. \quad (34)$$

Equality of the norms makes $H_a = H_b$ almost everywhere, and then everywhere by continuity. Their right derivatives are $-\{a > c\}$ and $-\{b > c\}$, respectively. Thus the distributions agree and so do the rearrangements. $\square$

Proof of the rigidity assertion in Theorem 1.1. Self-adjointness and the semigroup law give the half-time identities

$$Q_\Gamma(t) = \|w(t/2)\|_2^2, \quad Q'_\Gamma(t) = -\mathfrak{a}_\Gamma[w(t/2)]. \quad (35)$$

The identical formulas hold on the interval. Suppose $Q_\Gamma(T) = Q_I(T)$ for some $T > 0$. Theorem 1.2 at time $T/2$ gives $w(T/2) \preceq z(T/2)$. Lemma 5.2 and (35) imply

$$w(T/2)^* = z(T/2) \quad \text{almost everywhere on } (0, L). \quad (36)$$

The differentiable nonnegative function $Q_I(t) - Q_\Gamma(t)$ attains its minimum zero at the interior point T , so its derivative vanishes there. Thus (35) gives

$$\mathfrak{a}_\Gamma[w(T/2)] = \mathfrak{a}_I[z(T/2)] = \int_0^L |(w(T/2)^*)'|^2. \quad (37)$$

Set $u = w(T/2)$ and $M = \max u$. By Lemma 5.1, only finitely many critical or vertex values need be excluded. At all other levels $c \in (0, M)$ use the notation $a(c), b(c), N(c)$ from (19). Lemma 2.1 and one-dimensional coarea yield

$$\mathfrak{a}_\Gamma[u] = \int_0^M a(c) dc, \quad \int_0^L |(u^*)'|^2 = \int_0^M \frac{1}{b(c)} dc. \quad (38)$$

The second formula follows from $s = \mu(c)$, $ds = -b(c)dc$ and $(u^*)'(\mu(c)) = -1/b(c)$ on every interval between exceptional values. The Lipschitz representative ensures that these identities lose no singular energy. By (22),

$$a(c) - \frac{1}{b(c)} \geq \frac{N(c)^2 - 1}{b(c)} \geq 0. \quad (39)$$

Equations (37)-(39) force $N(c) = 1$ for almost every regular c .

Every edge incidence at a Dirichlet vertex has strictly positive derivative into its edge. Indeed, the function is positive inside that edge, zero at its endpoint, and strictly concave; the endpoint derivative is at least any positive chord slope. Therefore, for all sufficiently small positive regular levels,

$$N(c) \geq \sum_{p \in D} \deg(p). \quad (40)$$

The small incident edge neighborhoods can be chosen disjoint, including the two incidences of a loop. It follows that D consists of one vertex and that this vertex has degree one.

Next consider a standard vertex p of degree $d \geq 3$. Write $c = u(p) > 0$ and let d_i be the derivatives along its outgoing incidences. Kirchhoff gives $\sum_i d_i = 0$. Each incidence with $d_i > 0$ gives increasing values immediately away from p ; each incidence with $d_i < 0$ gives decreasing values. If $d_i = 0$, strict edgewise concavity also gives decreasing values. Thus every incidence is assigned to one of two classes. Since $d \geq 3$, one class contains at least two incidences. On the corresponding side of c , all sufficiently close regular levels have at least two distinct preimages. This contradicts $N = 1$ almost everywhere. No standard vertex can have degree at least three.

A connected finite graph whose degrees are at most two and which has a vertex of degree one is a path. Its unique Dirichlet vertex is an endpoint; its other endpoint is standard and hence Neumann. Suppressing standard degree-two vertices changes neither the metric space nor its Laplacian. This proves necessity. Sufficiency follows by reflection and subdivision invariance of the interval problem. $\square$

The equality argument does not infer strictness from strict resolvent inequalities followed by a limit. Such a limit could erase a positive defect. Instead, equality at the specified time produces the exact energy equality (37).

6. Consequences and limits

Corollary 6.1 (convex temperature functionals). Under the hypotheses of Theorem 1.2, for every convex nondecreasing function $\Phi : [0, \infty) \rightarrow [0, \infty)$ with $\Phi(0) = 0$,

$$\int_{\Gamma} \Phi(S_{\Gamma}(t)f) dx \leq \int_0^L \Phi(S_I(t)g) dx. \quad (41)$$

In particular, all L^p norms, $1 \leq p \leq \infty$, are bounded by those of the comparison evolution.

Proof. Concentration order implies the stop-loss comparison (33) and the comparison of first moments. Every such convex Φ is a nonnegative linear function plus an integral of $(r - c)_+$ against its distributional second derivative; approximation applies if necessary. Integrating that representation proves (41). Choosing $\Phi(r) = r^p$ gives finite p ; passage to the limit gives $p = \infty$. At positive time the profiles are bounded by spectral smoothing and the edge Sobolev embedding. $\square$

The interval bound has the explicit expansion

$$Q_I(t) = \frac{8L}{\pi^2} \sum_{k=0}^{\infty} \frac{\exp\left(-\frac{(2k+1)^2\pi^2 t}{4L^2}\right)}{(2k+1)^2}. \quad (42)$$

Indeed, the normalized eigenfunctions are $\sqrt{2/L} \cos((2k+1)\pi x/(2L))$. Their squared integrals are $8L/(\pi^2(2k+1)^2)$, giving (42) by the spectral representation of (1).

Corollary 6.2 (positive time weights). Let $\rho : (0, \infty) \rightarrow [0, \infty)$ be measurable. Whenever the interval integral is finite,

$$\int_0^{\infty} \rho(t) Q_{\Gamma}(t) dt \leq \int_0^{\infty} \rho(t) Q_I(t) dt. \quad (43)$$

If ρ is positive on a set of positive measure, equality characterizes the same interval. In particular, choosing $\rho = 1$ recovers the sharp torsional-rigidity bound $\int_0^{\infty} Q_{\Gamma}(t) dt \leq L^3/3$.

Proof. Integrate Theorem 1.1 and use its strictness. For the last constant, the interval torsion function solves $-v'' = 1$, $v'(0) = v(L) = 0$, hence $v(x) = (L^2 - x^2)/2$ and $\int_0^L v = L^3/3$. $\square$

The torsional comparison itself is prior work; it is discussed in [BT, Remark 2.3] and is not a separate novelty claim here. Similarly, with the diffusion convention that the generator is d^2/dx^2 on each edge, $Q_{\Gamma}(t)/L$ is the survival probability at time t for the killed graph diffusion started from uniform length measure. Theorem 1.1 therefore also says that its lifetime is stochastically dominated by the lifetime on the mixed interval. This is the direct probabilistic interpretation of the heat-content bound, using the Feynman-Kac representation in [BT, Proposition 3.2].

The conclusions of Sections 1-6 concern unweighted finite metric graphs with the stated Dirichlet and Kirchhoff conditions. Section 7 gives a separate theorem for nonnegative killing measures with continuous form domain. They do not provide a heat-trace bound, pointwise heat-kernel domination,

or a theorem for arbitrary vertex couplings. We prove no quantitative stability estimate. A useful next problem is to quantify the loss in (2) in terms of geometric distance from a path, with explicit control of the time scale and of edges whose lengths tend to zero. Such a statement requires additional work beyond the qualitative rigidity proved here.

7. A fixed amount of nonnegative killing

The comparison extends to a different boundary regime: there are no Dirichlet conditions, and killing is described by a finite nonnegative measure. Throughout this section, Γ is a finite compact connected metric graph of total length L , and ν is a nonnegative Radon measure on Γ with

$$\nu(\Gamma) = \kappa > 0. \quad (44)$$

The measure ν is distinct from the length measure dx used for heat content and rearrangement. In particular, ν may charge vertices. On the space $H^1(\Gamma)$ of continuous edgewise H^1 functions, define

$$\mathfrak{a}_\nu[u] = \int_\Gamma |u'|^2 dx + \int_\Gamma |u|^2 d\nu, \quad S_\nu(t) = e^{-tA_\nu}, \quad Q_\nu(t) = \langle 1, S_\nu(t)1 \rangle. \quad (45)$$

Here A_ν is the operator associated with the form. This includes nonnegative integrable potentials and nonnegative delta interactions, separately or together, as well as singular continuous killing measures. It does not include signed potentials or weighted continuity conditions.

The comparison form on $I_L = (0, L)$ is

$$\mathfrak{a}_{I,\kappa}[v] = \int_0^L |v'|^2 dx + \kappa |v(L)|^2, \quad \mathcal{D}(\mathfrak{a}_{I,\kappa}) = H^1(0, L). \quad (46)$$

Its operator has $v'(0) = 0$ and $v'(L) + \kappa v(L) = 0$. Write $S_{I,\kappa}$ and $Q_{I,\kappa}$ for its semigroup and heat content.

Theorem 7.1 (fixed-total-killing comparison and rigidity). Suppose ν satisfies (44), $0 \leq f \in L^2(\Gamma)$, and $0 \leq g \in L^2(0, L)$ is nonincreasing with $f \preceq g$. Then $S_{I,\kappa}(t)g$ is nonnegative and nonincreasing, and

$$\mathcal{C}_{S_\nu(t)f}(s) \leq \int_0^s S_{I,\kappa}(t)g(r) dr \quad (0 \leq s \leq L, t > 0). \quad (47)$$

In particular,

$$Q_\nu(t) \leq Q_{I,\kappa}(t) \quad (t > 0). \quad (48)$$

Equality in (48) at one positive time holds if and only if Γ is a path of length L and $\nu = \kappa \delta_p$ at one endpoint p . Subdivision and reflection are immaterial. Thus the interval with all killing concentrated at one endpoint is the unique maximizer, as a metric space with killing measure, among the stated graphs and measures with fixed L and κ .

For killing supported on finitely many vertices, the time integral of (48) is the already known sharp torsion inequality of Özcan and Täufer [OT, Theorem 8.2 in the cited preprint]. Their proof already places all strength at a minimum and uses energy rearrangement. Our additional assertion is the all-times concentration comparison (47), its extension to arbitrary finite measures, and rigidity from equality at one time. The key extra ingredient is a boundary inequality for the cumulative resolvent, rather than an energy comparison alone. Generalized couplings are also discussed in [BM, Remark 4.11]; that discussion

does not supply (47). The fixed-graph ground-state maximization problem of Kurasov and Serio [KS] concerns a different functional and optimization direction.

7.1. Form and interval preliminaries

The compact embedding $H^1(\Gamma) \hookrightarrow C(\Gamma)$ makes the measure term in (45) continuous on H^1 . The form norm after adding $\|u\|_2^2$ is equivalent to the H^1 norm, so the form is closed and has compact resolvent. It is a Dirichlet form: real normal contractions decrease both terms. Its semigroup and resolvents are therefore positivity preserving and sub-Markovian.

The spectrum is strictly positive. Indeed, choose $x_0 \in \Gamma$ with $|u(x_0)|^2 \leq \kappa^{-1} \int |u|^2 d\nu$, using continuity and the minimum of $|u|$. A simple path of length at most L from x_0 to x gives

$$\|u\|_2^2 \leq 2L^2 \int_{\Gamma} |u'|^2 dx + \frac{2L}{\kappa} \int_{\Gamma} |u|^2 d\nu. \quad (49)$$

In this section Lemma 2.1 has the following version without Dirichlet vertices: a continuous edgewise K -Lipschitz function has a K -Lipschitz decreasing rearrangement with $u^*(L) = \min_{\Gamma} u$. Use a simple path from a maximum to a minimum in the proof of (11); only levels between these two values are needed.

Fix $h > 0$ and let $B_{h,\kappa} = (1 + hA_{I,\kappa})^{-1}$. For nonnegative nonincreasing g , the function $v = B_{h,\kappa}g$ is nonnegative and nonincreasing. For smooth g , its derivative satisfies (14), now with $q(L) = -\kappa v(L) \leq 0$; the same positive-part argument applies. Monotone smooth approximation proves the assertion for L^2 data. For $V(s) = \int_0^s v$ and $G(s) = \int_0^s g$ one has

$$-hV'' + V = G, \quad V(0) = 0, \quad V(L) + h\kappa V'(L) = G(L). \quad (50)$$

The differential equation follows by integration from the Neumann endpoint, and its value at L , together with $v'(L) = -\kappa v(L)$, gives the last condition. This is a Robin condition for V with an inhomogeneous boundary value determined by the full integral of the datum.

We shall use the following elementary comparison. If $W \in H^2(0, L)$ satisfies

$$-hW'' + W \leq 0, \quad W(0) = 0, \quad W(L) + h\kappa W'(L) \leq 0, \quad (51)$$

then integration against W_+ gives

$$h \int_0^L |(W_+)'|^2 + \int_0^L W_+^2 + \frac{W_+(L)^2}{\kappa} \leq 0. \quad (52)$$

Thus $W \leq 0$. In particular, (50) preserves cumulative order of decreasing data: if $G_1 \leq G_2$ on $[0, L]$, their solutions satisfy $V_1 \leq V_2$, including when $G_1(L) < G_2(L)$.

7.2. Resolvent comparison for finitely many killing points

First suppose $\nu = \sum_p \beta_p \delta_p$, with finitely many $\beta_p \geq 0$ and $\sum_p \beta_p = \kappa$. Insert degree-two vertices at interior atoms. The operator acts as $-u''$ on each open edge and satisfies

$$\sum_{e \sim p} \partial_{\text{out},e} u(p) = \beta_p u(p). \quad (53)$$

Here each derivative points from the vertex into its incident edge; this convention makes the derivative at the right endpoint of an interval equal to $-u'(L)$. Vertices without an atom have $\beta_p = 0$.

Take a strictly positive nonconstant polynomial load on each closed edge, and put $u = (1 + hA_\nu)^{-1}f$. As in Lemma 3.2, u is analytic and nonconstant on each edge. It is strictly positive everywhere. An interior zero contradicts $-hu'' + u = f > 0$. At a zero vertex all outgoing derivatives are nonnegative and sum to zero by (53); each is zero, and $u''(p) = -f(p)/h < 0$ into each edge again contradicts nonnegativity. Thus $m := \min u > 0$.

For a level $c \in (m, \max u)$ outside the finite set of critical and vertex values, use E_c, μ, N, a, b as in (19). There is at least one level preimage, by a path from a maximum to a minimum. Integrating over the superlevel set, with the vertex condition (53), gives

$$h \left(a(c) + \sum_{u(p) > c} \beta_p u(p) \right) + U(\mu(c)) = \int_{E_c} f \leq F(\mu(c)), \quad (54)$$

where $U = \mathcal{C}_u$ and $F = \mathcal{C}_f$. In particular, the vertex term has a positive sign. Discard it and use $ab \geq N^2 \geq 1$ and $U''(\mu(c)) = -1/b(c)$ to obtain

$$-hU'' + U \leq F \quad \text{almost everywhere on } (0, L). \quad (55)$$

The rearrangement is Lipschitz by Section 7.1, and only finitely many ranks are exceptional; thus this is also a weak inequality. The new boundary information follows by testing the full resolvent equation against the constant function:

$$U(L) + h \sum_p \beta_p u(p) = F(L), \quad U'(L) = m. \quad (56)$$

Since every $u(p) \geq m$, (56) implies

$$U(0) = 0, \quad U(L) + h\kappa U'(L) \leq F(L). \quad (57)$$

Equations (55) and (57), compared to (50) with datum f^* , give $U \leq V$ by (51)-(52). The polynomial density argument and the rearrangement contraction (10) pass this inequality to every nonnegative L^2 load. The last assertion following (52) then gives

$$\mathcal{C}_{(1+hA_\nu)^{-1}f}(s) \leq \int_0^s B_{h,\kappa} g(r) dr \quad \text{whenever } f \preceq g \quad (58)$$

for nonnegative f and nonnegative decreasing g .

7.3. General measures and evolution

Choose finite nonnegative atomic measures ν_j converging weakly to ν , all with mass κ ; finite partitions into sets of diameter tending to zero give such measures. On the fixed graph, the set of products uv with $\|u\|_{H^1}, \|v\|_{H^1} \leq 1$ is relatively compact in $C(\Gamma)$. Weak convergence and the uniform bound $\|\nu_j - \nu\|_{\text{TV}} \leq 2\kappa$ therefore imply

$$\epsilon_j := \sup_{\|u\|_{H^1}, \|v\|_{H^1} \leq 1} \left| \int_\Gamma uv d(\nu_j - \nu) \right| \longrightarrow 0. \quad (59)$$

For clarity, take a finite uniform ϵ -net of the compact closure of that product set. The integrals tend to zero on the net, and the error on the whole set is at most $2\kappa\epsilon$. Then let ϵ tend to zero.

The bilinear forms $\langle u, v \rangle + h\mathfrak{a}_{\nu_j}(u, v)$ are uniformly coercive on H^1 , with coercivity constant at least $\min(1, h)$ for the usual norm. Subtracting their weak resolvent equations, testing with the difference, and applying (59) gives

$$(1 + hA_{\nu_j})^{-1}f \longrightarrow (1 + hA_\nu)^{-1}f \quad \text{in } H^1(\Gamma) \quad (60)$$

for every $f \in L^2(\Gamma)$. For example, if u is the limiting resolvent solution, the norm of the difference is bounded by $h\epsilon_j\|u\|_{H^1}/\min(1, h)$. Equation (9) now passes (58) to ν .

Iterate this resolvent comparison with $h = t/n$. All interval iterates stay decreasing and preserve cumulative order by (50)-(52). Spectral dominated convergence, exactly as in (28), gives (47). Taking $f = g = 1$ gives (48). This proves the inequality part of Theorem 7.1 without any regular-level assumption for the heat flow or for a singular measure.

7.4. Equality for an arbitrary killing measure

We give the additional details needed because strict edgewise concavity is generally false in the presence of a distributed potential.

First, every nonnegative $u \in H^1(\Gamma)$ satisfies the classical rearrangement inequality

$$u^* \in H^1(0, L), \quad \int_\Gamma |u'|^2 dx \geq \int_0^L |(u^*)'|^2 ds. \quad (61)$$

One direct extension of the proof in Section 5 is to approximate u in H^1 by nonnegative continuous piecewise linear functions on the edges. At every nonexceptional value their energy coarea formulas and $ab \geq N^2 \geq 1$ prove (61); their finitely many constant pieces contribute zero energy on both sides. Rearrangement is continuous in L^2 by (10). The rearranged approximants are bounded in H^1 , so weak compactness and lower semicontinuity pass the estimate to u . Continuity and connectedness also give $u^*(L) = \min u$ whenever the rearrangement has its continuous representative. Hence

$$\mathfrak{a}_\nu[u] \geq \int_0^L |(u^*)'|^2 ds + \kappa(\min u)^2 = \mathfrak{a}_{I,\kappa}[u^*]. \quad (62)$$

Suppose now that equality in (48) holds at $T > 0$. Set $u = S_\nu(T/2)1$ and $z = S_{I,\kappa}(T/2)1$. The half-time identities (35), with the corresponding forms, and Lemma 5.2 give $u^* = z$. Differentiating the nonnegative heat-content deficit at its zero gives

$$\mathfrak{a}_\nu[u] = \mathfrak{a}_{I,\kappa}[z]. \quad (63)$$

The interval profile z is strictly positive on $[0, L]$ and strictly decreasing on $(0, L)$. To see strictness, the maximum-principle argument in Lemma 5.1 applies to the interval with its finite Robin endpoint: a zero there would have zero outward derivative and contradict the boundary point lemma. The semigroup is positivity improving, is sub-Markovian, and has strictly positive spectrum by (49). Thus $A_{I,\kappa}S_{I,\kappa}(t)1 > 0$ on the open interval as in (31)-(32). Consequently $z'' < 0$ and $z'(0) = 0$, so $z'(s) < 0$ for $s > 0$. In particular, its distribution has no atoms, including at its endpoint values.

Equations (62)-(63) force equality separately in the gradient and killing terms:

$$\int_\Gamma |u'|^2 = \int_0^L |z'|^2, \quad \int_\Gamma (u^2 - m^2) d\nu = 0, \quad m = \min u = z(L) > 0. \quad (64)$$

Thus ν is supported on the minimum set of u . We next show that this set is a single endpoint of a path.

Spectral smoothing gives $u \in \mathcal{D}(A_\nu) \subset H^1(\Gamma)$. On each open edge the weak equation is

$$u'' = u\nu - (A_\nu u) dx \quad (65)$$

as distributions, with the measure restricted to that edge. The right side is a finite signed measure. Thus u' has a representative of bounded variation and is bounded on the edge; u is edgewise Lipschitz. This assertion does not assert a sign for u'' .

Put $M = \max u = z(0)$. Since $u^* = z$, its distribution $\mu(c) = |\{u > c\}|$ equals $z^{-1}(c)$ for $m < c < M$, and is continuously differentiable there with $-\mu'(c) > 0$. The one-dimensional area formula now gives, at almost every such value, finite level multiplicity $N(c)$ and

$$b(c) = \sum_{u=c} |u'|^{-1} = -\mu'(c), \quad \int_\Gamma |u'|^2 = \int_m^M a(c) dc, \quad \int_0^L |z'|^2 = \int_m^M \frac{dc}{b(c)}. \quad (66)$$

Here we may exclude vertex values, images of nondifferentiability points and images of points where $u' = 0$. These images have measure zero: Lipschitz maps preserve null sets, and the area formula makes the image of the zero-derivative set null. Moreover that set itself has length zero in this case. Its pushforward is supported on a null set of levels, whereas the distribution of u , being that of z , is absolutely continuous on (m, M) and gives no mass to the endpoint levels. Thus the reciprocal-derivative formula loses no critical-set mass. Finite multiplicity almost everywhere follows from $\int N(c)dc = \int_\Gamma |u'|dx < \infty$. These facts justify (66) without a finite-critical-points assertion.

The gradient equality in (64) and $ab \geq N^2$ yield

$$0 = \int_m^M \left(a(c) - \frac{1}{b(c)} \right) dc \geq \int_m^M \frac{N(c)^2 - 1}{b(c)} dc \geq 0. \quad (67)$$

Hence $N(c) = 1$ for almost every $c \in (m, M)$. No edge subinterval is constant, because that would create an atom in the distribution of u . At a vertex of degree at least three choose disjoint small neighborhoods of its incidences. On each neighborhood u takes a value different from the vertex value. By continuity, each incidence therefore covers an open interval of values either above or below the vertex value. Two of the three incidences cover overlapping intervals on the same side. This produces at least two distinct preimages on an open interval of levels, contrary to (67). All vertex degrees are at most two.

A finite connected graph with these degrees is a path or a cycle. On a cycle, two arcs from a minimum to a maximum give at least two preimages of every intermediate value, so a cycle is impossible. On a path, a continuous function with no constant subinterval and almost everywhere single level multiplicity must be strictly monotone: a failure of monotonicity produces a strict rise followed by a fall, or a strict fall followed by a rise, giving two preimages on an open interval of values. Its minimum is therefore attained at one endpoint only. The support conclusion from (64) gives $\nu = \kappa\delta_p$ there. Conversely, this measure on the path is the comparison problem, proving equality. Theorem 7.1 follows. $\square$

Corollary 7.2 (integrated comparison). Under (44),

$$\int_0^\infty Q_\nu(t) dt \leq \frac{L^3}{3} + \frac{L^2}{\kappa}, \quad (68)$$

with equality exactly in the case of Theorem 7.1. More generally, the positive-time-weight and convex-temperature conclusions of Section 6 hold with the Robin comparator.

Proof. The positive spectral gap makes the unweighted time integral finite. The interval torsion function is $v(x) = (L^2 - x^2)/2 + L/\kappa$, whose integral is the right side of (68). Integrate (48) and use its strictness; the other conclusions follow as in Section 6. $\square$

For atomic vertex measures, (68) and its equality case are precisely prior work [OT, Theorem 8.2], not additional new results. Theorem 7.1 gives their all-times strengthening and allows general measures. The case $\kappa = 0$ is excluded: every graph then retains the constant temperature and has heat content L , so rigidity fails. Quantitative stability and arbitrary vertex couplings remain outside the result. The two separate extremal comparisons do not settle the product problem discussed in [OT, Section 8.3]. Section 8 supplies the required joint argument for arbitrary finite nonnegative killing measures.

8. Kohler-Jobin inequalities for general killing measures

Let Γ be a finite compact connected metric graph of length $L > 0$, without Dirichlet conditions, and let ν be any finite nonnegative Borel measure of mass $\kappa > 0$. The form domain is the continuous space $H^1(\Gamma)$, as in Section 7. Set

$$\lambda = \lambda_1(A_\nu), \quad T = \int_{\Gamma} A_\nu^{-1} 1 \, dx, \quad s = \kappa L. \quad (69)$$

For $z > 0$, let $\vartheta(z) \in (0, \pi/2)$ solve $\vartheta \tan \vartheta = z$. The interval I_L with Neumann condition at 0 and killing $\kappa \delta_L$ has eigenvalue $\vartheta(s)^2/L^2$ and torsion $L^3/3 + L^2/\kappa$.

Theorem 8.1 (fixed length and total killing). Under these hypotheses,

$$\lambda T^{2/3} \geq \vartheta(\kappa L)^2 \left(\frac{1}{3} + \frac{1}{\kappa L} \right)^{2/3}. \quad (70)$$

Equality holds if and only if Γ is a path and ν is concentrated at one endpoint, up to subdivision and reflection.

Theorem 8.2 (fixed torsion and total killing). Let $\ell > 0$ satisfy $\ell^3/3 + \ell^2/\kappa = T$. Then

$$\lambda \geq \lambda_1(A_{I_\ell, \kappa}). \quad (71)$$

Equivalently, with $b = \kappa/\sqrt{\lambda}$ and $y = \arctan b$,

$$\lambda^{3/2} T \geq \frac{y^3}{3} + \frac{y^2}{b}. \quad (72)$$

The equality case is the same as in Theorem 8.1. Both theorems include point interactions, integrable nonnegative potentials and singular continuous killing measures.

These results give sharp fixed-constraint formulations of the Robin/delta joint-optimization direction discussed in [OT, Section 8.3 of the cited preprint], extending the one-point statements of the preceding revision. The Dirichlet graph Kohler-Jobin theorem is due to Mugnolo and Plümer [MP, Theorem 5.8]; its nonlinear Dirichlet extension is considered in [O26]. Modified-torsion and level-set methods are established tools in this subject. Our proof uses a ground-state distribution functional in which Cauchy-Schwarz absorbs all killing terms. It then compares a scalar differential inequality with the cosine profile. Eigenfunction distribution inequalities of this kind belong to the classical Chiti comparison method; [DS, Lemma 3.2] treats Robin domains on the range of interior superlevel sets. Here connectedness and nonnegative killing give (79) over the full range of regular levels. The Euclidean domain Robin questions in [BCS, Section 5] impose different constraints and are not resolved here.

8.1. A ground-state distribution estimate

Define the continuous increasing function

$$B(x) = \frac{x^3}{3} + x^2 \cot x \quad (0 < x < \pi/2), \quad B(0) = 0, \quad B(\pi/2) = \frac{\pi^3}{24}. \quad (73)$$

Proposition 8.3. For every graph and measure above, with $k = \sqrt{\lambda}$,

$$T \geq k^{-3} B(\min\{kL, \pi/2\}). \quad (74)$$

Proof for finitely supported measures. Subdivide the graph at the atoms and write $\nu = \sum_p \beta_p \delta_p$, where $\beta_p \geq 0$. Choose a positive first eigenfunction ψ , and put $m = \min \psi > 0$ and $M = \max \psi$. Positivity follows by taking the absolute value in the form minimization and applying edgewise uniqueness: at a zero minimum the nonnegative outgoing derivatives sum to zero, and propagation along the connected graph would give the zero function. On each open edge, $\psi'' = -\lambda\psi < 0$. Thus no edge piece is constant, there are finitely many critical and vertex values, and $m < M$.

For $0 \leq t < M$, define

$$\mu(t) = |\{\psi > t\}|, \quad H(t) = \lambda \int_{\{\psi > t\}} \psi \, dx, \quad g(t) = \int_0^t \frac{\mu(r)}{H(r)} \, dr. \quad (75)$$

The positive minimum gives $0 \leq g' \leq 1/(\lambda m)$, so g extends to a Lipschitz function on $[0, M]$, with $g(0) = 0$. Set $G(t) = \int_0^t g'(r)^2 \, dr$. Cauchy-Schwarz gives $g(t)^2 \leq tG(t)$. Testing the weak eigenvalue equation with $G(\psi)$ therefore yields

$$\begin{aligned} \mathfrak{a}_\nu[g(\psi)] &= \int_\Gamma g'(\psi)^2 |\psi'|^2 \, dx + \int_\Gamma g(\psi)^2 \, d\nu \\ &\leq \int_\Gamma g'(\psi)^2 |\psi'|^2 \, dx + \int_\Gamma \psi G(\psi) \, d\nu \\ &= \lambda \int_\Gamma \psi G(\psi) \, dx = \int_0^M H(t) g'(t)^2 \, dt. \end{aligned} \quad (76)$$

Every test function here belongs to $H^1(\Gamma)$. The chain rules hold almost everywhere; the finitely many critical points cause no ambiguity. Cavalieri's formula and the torsion variational principle now give

$$\begin{aligned} T &= \sup_{v \in H^1(\Gamma)} \left(2 \int_\Gamma v \, dx - \mathfrak{a}_\nu[v] \right) \\ &\geq \int_0^M (2\mu g' - H(g')^2) \, dt = \int_0^M \frac{\mu(t)^2}{H(t)} \, dt. \end{aligned} \quad (77)$$

In particular, killing away from a ground-state minimum is included in (76); it has not been dropped from the energy.

Let $q = \psi^*$ be the decreasing rearrangement on $[0, L]$, and put $U(s) = \int_0^s q(r) \, dr$. The function q is positive and Lipschitz, with $q(0) = M$ and $q(L) = m$. To justify Lipschitz continuity, let $D_0 = \max_\Gamma |\psi'|$. A path crossing two levels $t_1 < t_2$ spends length at least $(t_2 - t_1)/D_0$ between them. Hence $\mu(t_1) - \mu(t_2) \geq (t_2 - t_1)/D_0$, which gives the asserted inverse bound. All level and inverse substitutions can also be made on the finitely many intervals of regular values and then passed to their endpoints.

For a regular $t \in (m, M)$, flux integration gives

$$a(t) + \sum_{\psi(p) > t} \beta_p \psi(p) = H(t), \quad a(t) = \sum_{\psi=t} |\psi'|, \quad b(t) = \sum_{\psi=t} |\psi'|^{-1} = -\mu'(t). \quad (78)$$

Connectedness implies that a regular level has at least one preimage, so $a(t)b(t) \geq 1$. Since the killing coefficients are nonnegative, $a(t) \leq H(t)$. Using $H(t) = \lambda U(\mu(t))$ and inverting the distribution function gives

$$0 \leq -q'(s) \leq \lambda U(s) \quad \text{for almost every } s \in (0, L). \quad (79)$$

The interval $0 < t < m$ in (77) has $\mu(t) = L$ and $H(t) = \lambda U(L)$. Substitution on the remaining levels, followed by integration by parts, yields

$$\begin{aligned} \lambda \int_0^M \frac{\mu(t)^2}{H(t)} dt &= \frac{mL^2}{U(L)} + \int_0^L \frac{s^2}{U(s)} (-q'(s)) ds \\ &= \int_0^L \left(2 \frac{sq(s)}{U(s)} - \frac{s^2 q(s)^2}{U(s)^2} \right) ds. \end{aligned} \quad (80)$$

The term at L cancels exactly, and $s^2 q(s)/U(s) \rightarrow 0$ at zero because $U(s)/s \rightarrow M$. Including the interval below the positive minimum is essential.

Put $r(s) = U(s)/q(s)$, with $r(0) = 0$. Since q is decreasing, $r(s) \geq s$. Equation (79) gives

$$r'(s) = 1 - \frac{U(s)q'(s)}{q(s)^2} \leq 1 + k^2 r(s)^2, \quad \frac{d}{ds} \arctan(kr(s)) \leq k. \quad (81)$$

Consequently $r(s) \leq \tan(ks)/k$ for $0 < s < \min\{L, \pi/(2k)\}$. Thus $z(s) = s/r(s)$ lies in $(0, 1]$ everywhere and satisfies $z(s) \geq ks \cot(ks)$ on this initial interval. The function $2z - z^2$ is nonnegative and increasing for $0 \leq z \leq 1$. Combining (77)-(81) gives, with $a = \min\{kL, \pi/2\}$,

$$\begin{aligned} T &\geq k^{-2} \int_0^L (2z(s) - z(s)^2) ds \\ &\geq k^{-3} \int_0^a (2x \cot x - x^2 \cot^2 x) dx = k^{-3} B(a). \end{aligned} \quad (82)$$

The last identity follows by differentiation of (73). This proves (74) for finite atomic measures.

Passage to arbitrary measures. Choose finite atomic measures $\nu_j \rightarrow \nu$ of mass κ on the same graph. The compact embedding $H^1(\Gamma) \hookrightarrow C(\Gamma)$ and bounded total masses give

$$\epsilon_j := \sup_{\|u\|_{H^1}, \|v\|_{H^1} \leq 1} \left| \int_{\Gamma} uv d(\nu_j - \nu) \right| \longrightarrow 0. \quad (83)$$

Indeed, products of two unit-ball functions form a relatively compact subset of $C(\Gamma)$, so weak convergence is uniform on that set by a finite-net argument. The coercivity estimate in Section 7 is uniform in j . If $w_j = A_{\nu_j}^{-1}1$ and $w = A_{\nu}^{-1}1$, subtracting the form equations and testing with $w_j - w$ therefore gives $w_j \rightarrow w$ in H^1 and $T_j \rightarrow T$.

For completeness, the first eigenvalues converge as well. Testing with a normalized ground state of A_{ν} gives $\limsup \lambda_j \leq \lambda$. Conversely, normalized ground states for A_{ν_j} have uniformly bounded H^1 norms; a subsequence converges weakly in H^1 and strongly in C and L^2 . Its limit has norm one, and weak lower semicontinuity together with (83) gives $\lambda \leq \liminf \lambda_j$. Thus $\lambda_j \rightarrow \lambda > 0$. The continuity of the right side of (74) completes the proof. $\square$

8.2. Recovering the prescribed constraints and equality

We also need the familiar first-eigenvalue comparison

$$\lambda \geq \lambda_1(A_{I_L, \kappa}) = \frac{\vartheta(\kappa L)^2}{L^2}. \quad (84)$$

Here is a direct form proof that fixes the direction. For a nonnegative $u \in H^1(\Gamma)$, the graph energy rearrangement proved in Section 7 and the minimum bound give

$$\mathfrak{a}_\nu[u] \geq \int_0^L |(u^*)'|^2 ds + \kappa u^*(L)^2 \geq \lambda_1(A_{I_L, \kappa}) \|u\|_2^2. \quad (85)$$

Taking absolute values first handles arbitrary u , and minimization proves (84).

For $0 < x < \pi/2$, differentiation shows

$$B'(x) = x \cot x (2 - x \cot x) > 0, \quad (86)$$

because $0 < x \cot x < 1$. For Theorem 8.1, put $\theta = \vartheta(\kappa L)$. Equation (84) gives $kL \geq \theta$. Therefore (74) and strict monotonicity give

$$\lambda T^{2/3} \geq B(\min\{kL, \pi/2\})^{2/3} \geq B(\theta)^{2/3} = \theta^2 \left(\frac{1}{3} + \frac{1}{\kappa L} \right)^{2/3}. \quad (87)$$

If equality holds, $\theta < \pi/2$ and strict monotonicity force $kL = \theta$. Equality in (87) then gives $T = L^3/3 + L^2/\kappa$. Corollary 7.2 characterizes the equality case as the endpoint-killed interval. Conversely, that interval attains equality.

For Theorem 8.2, let $b = \kappa/k$ and $y = \arctan b$. If $kL \geq \pi/2$, (74) is strictly stronger than (72). If $kL < \pi/2$, (84) and monotonicity of $x \tan x$ imply

$$kL \tan(kL) \geq \kappa L, \quad kL \geq \arctan(\kappa/k) = y. \quad (88)$$

Hence (74) gives $k^3 T \geq B(y) = y^3/3 + y^2/b$, proving (72). The interval of length y/k with strength κ has first eigenvalue k^2 and torsion $k^{-3} B(y)$. Interval torsion is strictly increasing with length, whereas its first eigenvalue is strictly decreasing; the latter also follows from $k_\ell \ell = \arctan(\kappa/k_\ell)$. Thus (72) is equivalent to (71). Equality forces $kL = y$ and again $T = L^3/3 + L^2/\kappa$, giving the same rigidity by Corollary 7.2. This proves both theorems. $\square$

The two general-measure questions left open in revision 1.2 are therefore settled by (70)-(72). Quantitative stability of these inequalities and of the heat-content comparison remains open. Section 9 extends the stationary joint comparisons to nonlinear killing. Signed killing measures, discontinuous vertex couplings, weighted edge forms and the Euclidean Robin domain problem are outside the present assertions.

9. Nonlinear killing and the p -Laplacian

The distribution argument extends to the full range $1 < p < \infty$. Fix a finite compact connected metric graph Γ of length L , without Dirichlet conditions, and a finite nonnegative Borel measure ν with $\nu(\Gamma) = \kappa > 0$. On the continuous space $W^{1,p}(\Gamma)$ set

$$E_{p,\nu}[u] = \int_\Gamma |u'|^p dx + \int_\Gamma |u|^p d\nu, \quad \lambda_p = \inf_{u \neq 0} \frac{E_{p,\nu}[u]}{\int_\Gamma |u|^p dx}. \quad (89)$$

Let w minimize $E_{p,\nu}[u]/p - \int_{\Gamma} u \, dx$, and distinguish its mass from the customary homogeneous torsional rigidity:

$$S_p = \int_{\Gamma} w \, dx, \quad T_p = S_p^{p-1}, \quad p' = \frac{p}{p-1}, \quad d_p = p' + 1. \quad (90)$$

This convention for T_p agrees with [O26]. In particular, the scale-invariant product is $\lambda_p S_p^{p/d_p} = \lambda_p T_p^{p/(2p-1)}$. The energy includes nonlinear point interactions as well as distributed killing. The results below concern stationary variational problems; no nonlinear heat-concentration theorem is asserted.

We specify the one-dimensional functions to avoid a normalization ambiguity. Define

$$A_p(t) = \int_0^t \frac{v^{p-2}}{1+v^p} \, dv, \quad a_p = A_p(\infty) = \frac{\pi}{p \sin(\pi/p)}, \quad \tau_p = A_p^{-1} : [0, a_p) \longrightarrow [0, \infty). \quad (91)$$

Thus $\tau_2 = \tan$. For $\sigma > 0$, let $\theta_p(\sigma) \in (0, a_p)$ be the unique solution of

$$\theta_p(\sigma) \tau_p(\theta_p(\sigma)) = \sigma^{1/(p-1)}. \quad (92)$$

Theorem 9.1 (nonlinear joint comparisons). For every $1 < p < \infty$ and every graph and measure above, put $\sigma = \kappa L^{p-1}$ and $\theta = \theta_p(\sigma)$. Then

$$\lambda_p S_p^{p/d_p} \geq (p-1) \theta^p \left(\frac{1}{d_p} + \sigma^{-1/(p-1)} \right)^{p/d_p}. \quad (93)$$

Furthermore, if $\ell > 0$ is chosen by

$$\frac{\ell^{d_p}}{d_p} + \ell^{p'} \kappa^{-1/(p-1)} = S_p, \quad (94)$$

then $\lambda_p \geq \lambda_p(I_{\ell}, \kappa \delta_{\ell})$. Equality in either comparison holds exactly when Γ is a path and all killing is at one endpoint, up to subdivision and reflection.

The nonlinear Dirichlet graph Kohler-Jobin inequality and its normalization are due to [O26, Theorem 5.2], following the nonlinear modified-torsion strategy of Kohler-Jobin and Brasco. Its level-set flux inequality and modified-torsion test are established ingredients [O26, Lemma 5.3]. Here the Hölder composition estimate below includes all finite killing terms, and the scalar comparison retains the finite Robin parameter. Thus the new assertion concerns nonnegative measure killing; the Dirichlet theorem is credited, not claimed anew.

9.1. Variational facts and interval rigidity

We first justify the variational objects and the separate interval comparisons. Path integration and Hölder's inequality give, uniformly over measures of mass κ ,

$$\|u\|_p^p \leq 2^{p-1} L^p \int_{\Gamma} |u'|^p \, dx + \frac{2^{p-1} L}{\kappa} \int_{\Gamma} |u|^p \, d\nu. \quad (95)$$

Indeed, $|u(x)| \leq |u(y)| + L^{1/p'} \|u'\|_p$; raise to the power p and integrate over x with length measure and over y with ν/κ . Consequently $E_{p,\nu}$ controls the $W^{1,p}$ norm. The compact embedding into $C(\Gamma)$, the direct method and strict convexity give a unique torsion minimizer $w \geq 0$, a positive minimum eigenvalue and a nonnegative normalized ground state. Strict convexity follows because equality in the

gradient term makes two functions differ by a constant, while the positive mass of ν excludes a nonzero constant difference.

The torsion Euler equation and its variational value are

$$\begin{aligned} \int_{\Gamma} |w'|^{p-2} w' v' dx + \int_{\Gamma} w^{p-1} v d\nu &= \int_{\Gamma} v dx, \\ (p-1)S_p &= \sup_v \left(p \int_{\Gamma} v dx - E_{p,\nu}[v] \right). \end{aligned} \quad (96)$$

Testing with w gives $E_{p,\nu}[w] = S_p$. The minimizer is not zero, since a small positive constant makes $E_{p,\nu}[u]/p - \int u$ negative. Optimizing scalar multiples of a test function also gives $T_p = \sup_{v \neq 0} (\int |v|^p)/E_{p,\nu}[v]$.

For the endpoint-killed interval, direct integration gives

$$w_I(x) = \frac{L^{p'} - x^{p'}}{p'} + \left(\frac{L}{\kappa} \right)^{1/(p-1)}, \quad S_{p,I} = \frac{L^{d_p}}{d_p} + L^{p'} \kappa^{-1/(p-1)}. \quad (97)$$

To compute its eigenvalue, put $c_p(x) = (1 + \tau_p(x)^p)^{-1/p}$. Differentiation of (91) gives $c'_p = -\tau_p c_p$ and

$$-(|c'_p|^{p-2} c'_p)' = (p-1)c_p^{p-1}, \quad c_p(0) = 1, \quad c'_p(0) = 0. \quad (98)$$

A positive interval ground state decreases from its Neumann endpoint; its flux equation and initial value therefore identify it, after normalization, with $c_p(kx)$, where $\lambda_p = (p-1)k^p$. Its Robin condition is $k^{p-1}\tau_p(kL)^{p-1} = \kappa$, so

$$\lambda_p(I_L, \kappa \delta_L) = (p-1) \frac{\theta_p(\kappa L^{p-1})^p}{L^p}. \quad (99)$$

Lemma 9.2 (separate comparisons). One has $\lambda_p \geq \lambda_p(I_L, \kappa \delta_L)$ and $S_p \leq S_{p,I}$. Equality in the torsion comparison holds exactly for the endpoint-killed interval.

Proof. The graph rearrangement inequality extends to p -energy:

$$\int_0^L |(u^*)'|^p ds \leq \int_{\Gamma} |u'|^p dx, \quad E_{p,\nu}[u] \geq E_{p,\kappa \delta_L}[u^*] \quad (u \geq 0). \quad (100)$$

For a continuous piecewise affine function, coarea and Hölder give $\alpha_p b^{p-1} \geq N^p \geq 1$, where $\alpha_p(t) = \sum_{u=t} |u'|^{p-1}$ and $b(t) = \sum_{u=t} |u'|^{-1}$. Integrating gives the gradient inequality. Piecewise affine approximation, the L^p contraction of rearrangement and weak lower semicontinuity extend it to $W^{1,p}$, including constant pieces. The killing term is at least $\kappa(\min u)^p$, and the continuous representative of u^* has endpoint value $\min u$. This proves (100). The Rayleigh and torsion variational principles give the two comparisons.

If $S_p = S_{p,I}$, uniqueness of the interval torsion minimizer gives $w^* = w_I$. Both nonnegative energy defects in (100) vanish, and ν is supported where $w = \min w$. We explain why the graph is a path even for singular measures. On each edge the distributional equation is

$$(|w'|^{p-2} w')' = w^{p-1} \nu - dx. \quad (101)$$

The flux is of bounded variation, hence bounded; therefore w' is bounded and w is Lipschitz. Its rearrangement w_I is strictly decreasing, with nonzero derivative on $(0, L]$. Its distribution is absolutely continuous between endpoint values and has no atoms. The one-dimensional area formula makes the

image of the critical and nondifferentiability set of w null. Absolute continuity of the distribution then shows that this set has zero length. Thus the coarea energy identities and inverse differentiation apply without losing mass, as in Section 7.4. Equality in (100) forces $N(t) = 1$ at almost every level.

There are no constant edge pieces, since they would create atoms in the distribution. At a vertex with at least three incidences, two disjoint incident germs have overlapping value intervals on the same side of the vertex value; those levels have at least two preimages. A cycle likewise has two arcs joining a minimum to a maximum. Hence the connected graph has no branch and no cycle, and is a path. A nonmonotone continuous function on a path would have repeated levels on an interval, so w is strictly monotone. Its minimum is one endpoint, where all of ν is concentrated. Formula (97) proves the converse. $\square$

9.2. The distribution bound with nonlinear killing

Define, continuously at the endpoints,

$$B_p(x) = \frac{x^{d_p}}{d_p} + \frac{x^{p'}}{\tau_p(x)} \quad (0 < x < a_p), \quad B_p(0) = 0, \quad B_p(a_p) = \frac{a_p^{d_p}}{d_p}. \quad (102)$$

Proposition 9.3. With $k = (\lambda_p/(p-1))^{1/p}$,

$$S_p \geq k^{-d_p} B_p(\min\{kL, a_p\}). \quad (103)$$

Proof for atomic measures. Subdivide at every atom. A nonnegative first eigenfunction ψ is strictly positive. Indeed, at a zero minimum every outgoing flux is nonnegative and their sum is zero. On an adjacent edge, the conserved quantity $(p-1)|\psi'|^p + \lambda_p \psi^p$ is then zero, forcing that edge solution to vanish; connectedness propagates this contradiction. The same argument excludes an interior zero. On each edge the flux derivative is $-\lambda_p \psi^{p-1} < 0$, so the derivative is strictly decreasing and there is at most one critical point. In particular there are no constant edge pieces. Write $m = \min \psi > 0$, $M = \max \psi$, and

$$\mu(t) = |\{\psi > t\}|, \quad H(t) = \lambda_p \int_{\{\psi > t\}} \psi^{p-1} dx, \quad g(t) = \int_0^t \left(\frac{\mu(r)}{H(r)} \right)^{1/(p-1)} dr. \quad (104)$$

Since $H \geq \lambda_p m^{p-1} \mu$, the function g is Lipschitz on $[0, M]$. Put $G(t) = \int_0^t g'(r)^p dr$. Hölder gives $g(t)^p \leq t^{p-1} G(t)$. Testing the weak eigenvalue equation with $G(\psi)$, and using the nonnegativity of ν , gives

$$\begin{aligned} E_{p,\nu}[g(\psi)] &\leq \int_{\Gamma} g'(\psi)^p |\psi'|^p dx + \int_{\Gamma} \psi^{p-1} G(\psi) d\nu \\ &= \lambda_p \int_{\Gamma} \psi^{p-1} G(\psi) dx = \int_0^M H(t) g'(t)^p dt. \end{aligned} \quad (105)$$

These compositions are in $W^{1,p}$ and all chain rules are justified by their Lipschitz bounds and the finite exceptional set. Equations (96) and (105), together with Cavalieri's formula, yield

$$S_p \geq \int_0^M \frac{\mu(t)^{p'}}{H(t)^{1/(p-1)}} dt. \quad (106)$$

Indeed the integrand before maximizing is $(p\mu g' - H(g')^p)/(p-1)$, and (104) attains its pointwise maximum.

Let $q = \psi^*$ and $U(s) = \int_0^s q(r)^{p-1} dr$. The path argument of Section 8 makes q positive and Lipschitz. On regular levels, total flux, including killing, equals H . Hölder's coarea inequality $\alpha_p b^{p-1} \geq 1$ therefore implies

$$0 \leq -q'(s) \leq (\lambda_p U(s))^{1/(p-1)} \quad \text{almost everywhere on } (0, L). \quad (107)$$

Including the levels $0 < t < m$ in (106), changing to the rank variable and integrating by parts gives

$$\begin{aligned} \lambda_p^{1/(p-1)} \int_0^M \frac{\mu(t)^{p'}}{H(t)^{1/(p-1)}} dt &= \frac{mL^{p'}}{U(L)^{1/(p-1)}} + \int_0^L \frac{s^{p'}}{U(s)^{1/(p-1)}} (-q'(s)) ds \\ &= \int_0^L F_p(z(s)) ds, \\ z(s) &= \left(\frac{sq(s)^{p-1}}{U(s)} \right)^{1/(p-1)}, \quad F_p(z) = \frac{pz - z^p}{p-1}. \end{aligned} \quad (108)$$

The term at L cancels the first term. The term at zero vanishes because $U(s)/s \rightarrow M^{p-1}$.

Put $r(s) = U(s)/q(s)^{p-1}$, so $r \geq s$, $r(0) = 0$ and $z = (s/r)^{1/(p-1)} \in (0, 1]$. From (107),

$$r' \leq 1 + (p-1)\lambda_p^{1/(p-1)} r^{p'} = 1 + ((p-1)kr)^{p'}. \quad (109)$$

For $R = (p-1)kr$, integration of $R'/(1+R^{p'}) \leq (p-1)k$ gives $\int_0^{R(s)} du/(1+u^{p'}) \leq (p-1)ks$. Substituting $u = v^{p-1}$ in this integral and using (91) shows that, for $ks < a_p$,

$$r(s) \leq \frac{\tau_p(ks)^{p-1}}{(p-1)k}, \quad z(s) \geq \frac{((p-1)ks)^{1/(p-1)}}{\tau_p(ks)}. \quad (110)$$

The right side belongs to $(0, 1]$: the model ratio U/q^{p-1} is at least the rank variable, or equivalently $\tau_p(x)^{p-1} \geq (p-1)x$, which follows directly from (91).

The function F_p is nonnegative and increasing on $[0, 1]$. Differentiating (102), using $\tau_p' = \tau_p^{2-p}(1+\tau_p^p)$, gives

$$B_p'(x) = \frac{p'x^{1/(p-1)}}{\tau_p(x)} - \frac{x^{p'}}{\tau_p(x)^p} = (p-1)^{-1/(p-1)} F_p\left(\frac{((p-1)x)^{1/(p-1)}}{\tau_p(x)}\right) > 0. \quad (111)$$

The endpoint values in (102) follow from (91), in particular $\tau_p(x) \sim ((p-1)x)^{1/(p-1)}$ as $x \downarrow 0$. Equations (106)-(111), with $a = \min\{kL, a_p\}$, imply

$$S_p \geq \lambda_p^{-1/(p-1)} \int_0^{a/k} F_p(z(s)) ds \geq k^{-d_p} \int_0^a B_p'(x) dx = k^{-d_p} B_p(a). \quad (112)$$

Only the interval before the pole of τ_p is compared. The remaining integrand, when present, is nonnegative. This proves the atomic case.

Passage to arbitrary measures. Approximate ν weakly on the fixed graph by finite atomic measures ν_j of the same mass κ . Compactness of the $W^{1,p}$ unit ball in $C(\Gamma)$ implies uniform convergence of $\int |u|^p d\nu_j$ to $\int |u|^p d\nu$ on bounded $W^{1,p}$ sets. The bound (95) is uniform in j . Testing Rayleigh quotients with a limiting ground state gives the eigenvalue limsup inequality; compactness of normalized minimizing ground states, convergence in C and weak lower semicontinuity of the gradient energy give the liminf inequality. Thus $\lambda_{p,j} \rightarrow \lambda_p > 0$.

For torsion, the minimizing values of $E_{p,\nu_j}/p - \int u$ are uniformly coercive by (95) and Young's inequality. Every subsequence of minimizers has a further subsequence converging weakly in $W^{1,p}$ and strongly in C . Lower semicontinuity and comparison with any fixed test function show that the limit minimizes $E_{p,\nu}/p - \int u$. Uniqueness identifies it with w , so $S_{p,j} \rightarrow S_p$. Continuity of the right side of (103), including at $kL = a_p$, completes the proof. $\square$

9.3. Proof of the joint comparisons

Lemma 9.2 and (99) give $kL \geq \theta$. Since B_p is strictly increasing,

$$\lambda_p S_p^{p/d_p} \geq (p-1)B_p(\min\{kL, a_p\})^{p/d_p} \geq (p-1)B_p(\theta)^{p/d_p}. \quad (113)$$

By (92), $B_p(\theta) = \theta^{d_p}(1/d_p + \sigma^{-1/(p-1)})$, so (113) is (93). Equality forces $kL = \theta < a_p$ and $S_p = S_{p,I}$; Lemma 9.2 gives the stated rigidity.

For the second comparison put $b = \kappa^{1/(p-1)}/k$ and $y = A_p(b)$. If $kL \geq a_p$, (103) gives $k^{d_p}S_p > B_p(y)$. Otherwise (99) and the monotonicity of $k\tau_p(x)$ imply $k\tau_p(kL) \geq \kappa^{1/(p-1)}$, hence $kL \geq y$. Therefore in all cases

$$k^{d_p}S_p \geq B_p(y) = \frac{y^{d_p}}{d_p} + \frac{y^{p'}}{b}. \quad (114)$$

The interval of length y/k with killing mass κ has eigenvalue $(p-1)k^p = \lambda_p$ and torsion mass $k^{-d_p}B_p(y)$. At fixed κ , (97) strictly increases with length, while the eigenvalue strictly decreases, as is also seen from $k\ell = A_p(\kappa^{1/(p-1)}/k)$. Thus (114) is equivalent to the comparison with the interval in (94). Equality forces $kL = y$ and again $S_p = S_{p,I}$, proving rigidity. The endpoint interval attains equality in both statements. $\square$

At $p = 2$, all definitions and constants reduce exactly to Section 8. As $\sigma \rightarrow \infty$, the sharp constant in (93) tends to $(p-1)a_p^p d_p^{-p/d_p}$, the nonlinear Dirichlet graph constant in [O26, equation (5.4)]. Quantitative stability, nonlinear heat concentration and signed-measure analogues remain outside these results.

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